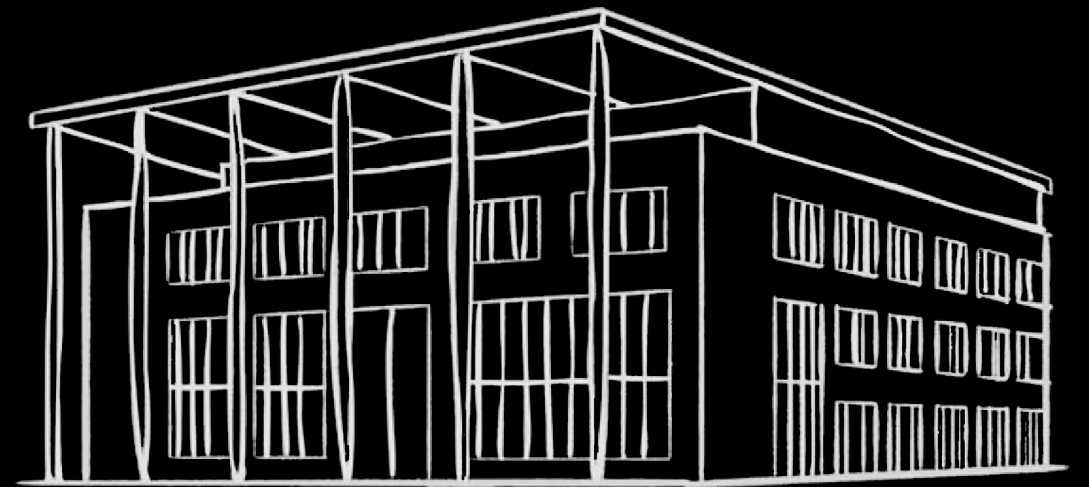


Coding Environments



**Q: What do you need
your code to do?**

**Q: How much code
do you want to see?**

Delegation

Verifiability

CODE AS DOCUMENTS

Notepad, TextEdit, Dropbox, Email

CODE AS TRACKED/REPRODUCIBLE

(Workflow paradigm)
git/GitHub, virtual environments, containers

INTEGRATED IDE

RStudio, VS Code, Cursor, Jupyter, PyCharm

AI-ASSISTED

GitHub Copilot, Cursor, Chat within IDE
Human driving line, or function, etc.

AGENTIC

Claude Code, Codex, Cursor Agent mode
SCHEMA.md

CODE AS DOCUMENTS

How do you identify the current version, and can you re-run last year's analysis today?

CODE AS TRACKED/REPRODUCIBLE

If you had to re-run your results today, what would break?

INTEGRATED IDE

What does your environment catch that you might miss?

AI-ASSISTED

How do you “check” AI-written code, and what does this mean for learning?

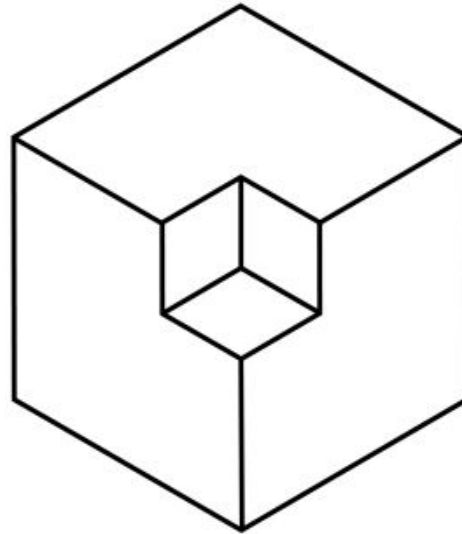
AGENTIC

What does it mean to be responsible for code you didn't write line by line?

**We tend to teach with
what we know best**

CODE

- Language (Python, R, etc.)
- Dependencies
- Versions
- Testing
- Build
- Delivery



COMPUTE

- Scale: Vertical and Horizontal
- Resources: CPU/mem/GPU
- Access to storage, DBs, APIs
- Requirements / provisioning
- Infrastructure as Code

DATA

- Source
- Velocity / Volume / Variety
- Cleaning
- Missing values /
- QA testing
- Versions
- Staging - into DB, file, API

